LINEAR REGRESSION REPORT

I. Dataset

The Mushroom dataset obtained from the UCI Machine Learning Repository, was used for the classification task using Logistic Regression [1]. It includes 8124 instances and 22 input features which are its physical characteristics that are predictors whether the Mushroom is poisonous(p) or edible(e).

II. Logistic Regression Result

Using the Scikit-learn library, a logistic regression model was used on the dataset in predicting whether the Mushroom is poisonous or edible.

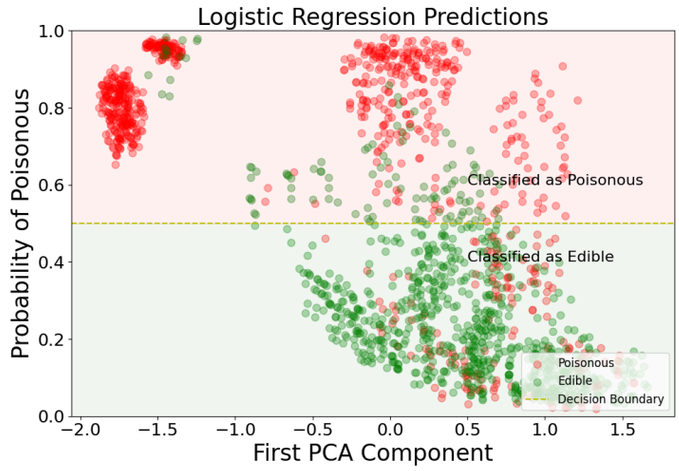


Figure 1: *Visualization of Logistic Regression Predictions*

The performance of the multivariate logistic regression model in predicting the mushroom characteristic is illustrated in a scatter plot against actual and predicted values. Green points are classified as edible while red points are classified as poisonous.

The model has adopted a proper strategy for making distinctions between the poisonous and edible samples with the help of PCA which is derived from the input features. The decision boundary that is positioned along the PCA axis is where predictions are made. For example, if a 0.8 probability of being poisonous resulted in classifying the sample as toxic, then it is possible to see how the model can handle high risk cases with precision.

This approach has sensitivity and specificity balanced while leveraging PCA for dimensionality reduction and a linear decision boundary for interpretability. The results indicate that the model is suitable for a number of applications which require accurate determination of potentially hazardous samples.

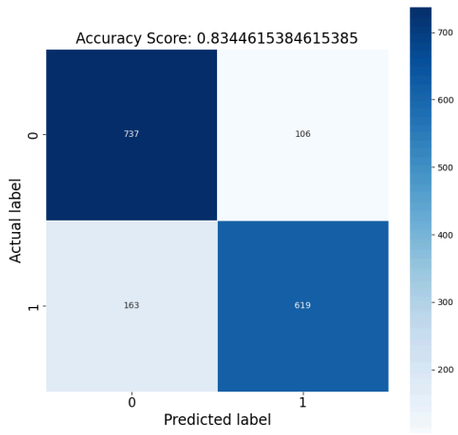


Figure 2: *Confusion matrix*

This **confusion matrix shows** the performance accuracy of a classification model as 83.45%. The model also correctly identified 737 true negatives and 619 true positives, while making wrong predictions of 106 false positives and 163 false negatives. The model performs a fairly good overall performance, however, there is room for enhancement especially in terms of reducing the false negatives or false positives depending on the concern of the application.

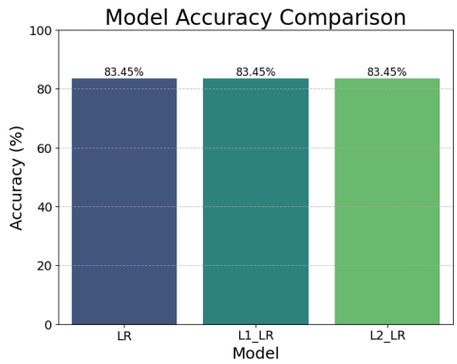


Figure 3: *Model and Regularization Comparison Results*

The comparison of model accuracies shows that the accuracies of three models that are Logistic Regression (LR), L1-regularized Logistic Regression (L1_LR) and L2-regularized Logistic Regression (L2_LR) is equal to 83.45% on the test set. This means that the use of L1 or L2 regularization did not affect the classification performance significantly. One possible reason could be that the data set might not be subject to overfitting thus regularization is not crucial in enhancing the generalization ability. Further, the feature set employed in training may have been previously optimized for instance the application ***Principle Component Analysis (PCA)***, hence leading to minimal changes across the different models.

III. References

[1] Home - UCI Machine Learning Repository. Retrieved February 19, 2025 from https://archive.ics.uci.edu/